

RPU-200

RADAR PROCESSING UNIT

MAINTENANCE MANUAL

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LEGACY SOURCE DOCUMENT - SYNTHETIC TRAINING MATERIAL

RECORD OF REVISIONS

Rev	Date	Description	Approved
A	2016-05-20	Initial issue.	J. W.
B	2017-09-04	Added RPU software update procedure and fault code list.	M. L.
C	2018-11-12	Added baseline 4.3 support, LAN update option and service-status checks.	R. K.

IMPORTANT INFORMATION

This manual was written for trained field service personnel. It combines equipment description, software maintenance, fault isolation and reference information in one document. Procedures may refer to menus or software that are not installed on every RPU-200 configuration.

WARNING: Hazardous voltages are present inside the RPU cabinet. Isolate external power before removing covers unless a procedure specifically requires power for measurement.

CAUTION: Never switch off power while processor firmware is being written. An incomplete update can make the processor unit unavailable.

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1. GENERAL INFORMATION

1.1 Scope

The RPU-200 is the processing computer used in the SRX family of surveillance radar systems. The unit receives digitized sensor data, performs plot and track processing, and distributes target information to operator display stations over the radar LAN. This manual gives information for installation, software servicing and first-line fault isolation of the Radar Processor.

1.2 Equipment covered

The instructions apply to RPU-200 units fitted with Processor Module PM-20 and Interface Module IF-12. Unless otherwise stated, the procedures apply to both Variant A and Variant B hardware. Variant B units have the dual-network IF-12B board and require software version 4.3.0 or later.

1.3 Maintenance philosophy

Faults are normally isolated by use of the internal service monitor and the operator station maintenance pages. Replaceable items are changed at line level. Board repair is not covered. Where software recovery does not correct a fault, replace the applicable line replaceable unit and return the removed item to depot maintenance.

1.4 Terms used in this manual

RPU	Radar Processing Unit / Radar Processor Unit
RMS	Radar Maintenance Software
RPS	Radar Processing Service
OSD	Operator System Display
LAN-A/LAN-B	Radar local area network channels
Service laptop	Approved maintenance PC

2. SAFETY PRECAUTIONS

2.1 General

Only trained personnel shall perform maintenance on the RPU-200. Prior to opening the cabinet, make sure that the radar has been removed from operational service and that the antenna drive cannot start unexpectedly.

WARNING: Disconnect the equipment power supply before removing the RPU top cover. Capacitors can retain voltage for a short period after power is removed.

WARNING: The radar may radiate RF energy when placed in an operational mode. Maintenance personnel should obey the site radiation safety instructions.

NOTE: Some software checks in Chapters 4, 7 and 8 require the RPU to remain powered. Do not remove covers for these checks.

2.2 Electrostatic discharge

Use an approved ESD wrist strap when handling PM-20, IF-12 or storage devices. Put removed boards in conductive protective packaging.

3. SYSTEM DESCRIPTION

3.1 Functional overview

The RPU consists of an industrial processor, solid-state system disk, radar interface board and two Ethernet interfaces. At startup the operating system loads the Radar Processing Service and supporting processes. After initialization, the RPU exchanges health and track data with the Operator System Display.

Item	Function	Normal indication
PM-20 Processor Module	Runs RPU operating system and processing software	STATUS LED green
IF-12 Interface Module	Radar I/O and timing interface	LINK green
System SSD	Stores OS, application and configuration	No front-panel indication
LAN-A	Primary radar network	Link/activity LEDs flashing
LAN-B	Secondary network, Variant B	Link/activity LEDs flashing

3.2 Operating modes

The radar system provides OPERATIONAL and MAINTENANCE selections at the operator station. Software installation should normally be carried out in MAINTENANCE mode. On early Variant A installations this selection is shown as SERVICE instead of MAINTENANCE.

4. SOFTWARE CONFIGURATION

4.1 Software packages

RPU software is supplied as an approved package on removable media or from the site maintenance repository. Each package includes the Radar Processing Service, display-interface components and a manifest file. Do not combine files from different approved packages.

Package	RPS	Display interface	Applicable hardware
RPU-4.2.7	4.2.7	3.8.4	Variant A
RPU-4.3.0	4.3.0	3.9.0	Variant A / Variant B
RPU-4.3.1	4.3.1	3.9.1	Variant A / Variant B
RPU-4.3.3	4.3.3	3.9.1	Variant B recommended

4.2 Check the installed version

1. At the Operator System Display, select RADAR > MAINTENANCE > SERVICE STATUS.
2. Select the RPU tab.
3. Read the RPS Version field and compare it with the software release record.
4. If the displayed version is not the approved version, update the RPU as described in Chapter 5.

NOTE: On software 4.1 and earlier, the version is shown under SYSTEM > ABOUT.

4.3 Configuration backup

Before major software changes the current site configuration should be saved. Insert an approved USB service device and use RMS > Configuration > Export. The exported file is named RPUCFG_<site>_<date>.zip. Store the file with the site service record.

5. UPDATING RPU SOFTWARE

5.1 Preparation

Use this procedure when an approved RPU software package must be installed. The RPU can be updated from an approved service laptop over the maintenance Ethernet connection. A USB update may be used when the network path is unavailable.

WARNING: Do not perform an update while the radar is required for an operational mission.

CAUTION: Do not disconnect the maintenance cable and do not remove power while the package is being installed.

5.2 Required equipment

- Approved service laptop with RMS 2.6 or later
- Ethernet service cable
- Approved RPU software package
- Current configuration record

5.3 Update through maintenance LAN

1. Set the radar system to MAINTENANCE mode.
2. Connect the service laptop to RPU connector J105 (SERVICE LAN).
3. Open Radar Maintenance Software.
4. Select RPU > Software > Package Management.
5. Select Import Package and browse to the approved RPU package.
6. Check that Package Status is VALID and that the package is applicable to the installed hardware.
7. Select Install.
8. Wait until the progress indicator reaches 100 percent. The RPU can restart automatically during this process.
9. When the message INSTALLATION SUCCESSFUL is shown, select Close.
10. At Service Status, verify the installed RPS version.
11. Record the new software version in the site configuration record.

NOTE: For Variant B, software 4.3.0 or later is required. Do not install package RPU-4.2.7 on a Variant B processor.

5.4 Update by USB device

If the service LAN is not available, copy the package to an approved USB device. Insert the device in the front service port and select RPU > Software > Local Media. Select the package and then proceed from step 6 above.

6. RESTORING THE PREVIOUS SOFTWARE VERSION

Use rollback only when the installed release has caused a verified operational or compatibility problem and the previous approved baseline is still authorized for the equipment configuration.

CAUTION: A configuration created by a later software version may not be compatible with an earlier release. Save the current configuration before rollback.

6.1 Roll back 4.3.x to 4.2.7

Applicability: Variant A only. Do not use this procedure on Variant B.

1. Set the radar to MAINTENANCE mode.
2. Export the current RPU configuration.
3. Connect the service laptop to J105.
4. Open RMS and select RPU > Software > Package Management.
5. Import package RPU-4.2.7.
6. Select Advanced and enable Allow downgrade.
7. Select Install and accept the downgrade message.
8. After the processor restarts, open Service Status.
9. Verify that RPS Version shows 4.2.7.
10. Import the last configuration known to be compatible with baseline 4.2.x.
11. Restart the processing service.
12. Verify that target data is received at the Operator System Display.

NOTE: If the processing service does not start after the rollback, see Fault Isolation 8.4.

6.2 Recovery after interrupted installation

If power was removed before installation completed, the standard package loader may not start. Connect the service laptop, hold the RECOVERY switch while applying power and use RMS Recovery > Restore Package. If the recovery loader cannot communicate with the RPU, replace the system SSD or PM-20 as directed by depot support.

7. SERVICE STATUS AND DIAGNOSTICS

7.1 Service Status page

The Service Status page shows the state of the software processes and network links. Select RADAR > MAINTENANCE > SERVICE STATUS. A green indicator shows normal operation. Yellow indicates initialization or degraded state. Red indicates a stopped process or a communication fault.

Item	Normal	Action if abnormal
Radar Processing Service	Running	See 8.2
Track Distributor	Running	See 8.3
LAN-A	Connected	Check cable and switch port
LAN-B	Connected (Variant B)	Check cable and switch port
Configuration	Loaded	Reload approved configuration

7.2 Diagnostic log

Select RMS > Diagnostics > Log Viewer. The most recent messages appear at the top. Use Export Log when information must be sent to engineering support. Do not edit exported logs.

7.3 Built-in test

Run the RPU self-test after board replacement or after a repeated processor fault. Select RMS > Diagnostics > Built-In Test and then select RUN ALL. The test takes approximately three minutes. A failed test displays a fault code. Refer to Appendix B.

8. FAULT ISOLATION

8.1 General

Before changing hardware, examine Service Status and the diagnostic log. Many faults are caused by loss of the radar LAN or an invalid software configuration. If a fault cannot be cleared by the actions below, record the fault code and contact system support.

8.2 Radar Processing Service not running

1. Open Service Status and confirm that Radar Processing Service is shown as Stopped or Failed.
2. Check that the installed RPS version agrees with the approved configuration record.
3. Select the service and click Restart.
4. Wait 30 seconds and check the status again.
5. If the service remains stopped, open the diagnostic log and record the first RPS error after the restart.
6. If fault code E217 is present, reload the approved configuration.
7. If fault code E231 is present, reinstall the current approved software package.
8. If the fault remains, replace the system SSD.

8.3 Operator display receives no target data

Check LAN-A first. If the link is down, examine the cable between J101 and the radar switch. On Variant B, verify LAN-B as well. If both links are up but no target data is available, restart the Track Distributor from Service Status. Should the fault remain, restart the RPU and repeat the check.

8.4 Fault after software rollback

If the RPU starts but the processing service is not running after rollback, verify that the imported site configuration was created for baseline 4.2.x. A 4.3 configuration should not be loaded into RPU-4.2.7. Restore the correct configuration and restart the processing service.

9. PREVENTIVE MAINTENANCE

9.1 Inspection interval

At six-month intervals, inspect the RPU for blocked air inlets, damaged network cables and abnormal fan noise. Verify that the installed software version agrees with the site configuration record.

9.2 Cleaning

Power off the RPU. Clean the external air filter with low-pressure dry air. Do not use solvent. If contamination is present inside the cabinet, refer the unit to depot maintenance.

9.3 Replace air filter

1. Set the radar out of service.
2. Switch off RPU power.
3. Remove the front air-filter cover.
4. Remove the filter element.
5. Install a clean filter element.
6. Install the cover.
7. Switch on the RPU and verify normal fan operation.

APPENDIX A - CONNECTOR REFERENCE

Connector	Function	Interface	Remarks
J101	Radar LAN-A	Ethernet	Primary network
J102	Radar LAN-B	Ethernet	Variant B only
J103	Timing input	Serial	Do not disconnect during operation
J104	Service serial	RS-232	Legacy recovery use
J105	Service LAN	Ethernet	Maintenance laptop
J106	USB service	USB 2.0	Approved media only
J201	Power input	28 VDC	External supply

APPENDIX B - SELECTED FAULT CODES

Code	Meaning	Initial action
E102	LAN-A link unavailable	Check J101 and network switch
E114	LAN-B link unavailable	Variant B: check J102
E217	Configuration load failed	Reload approved configuration
E231	RPS executable validation failed	Reinstall approved software package
E305	System SSD read error	Replace system SSD
E411	Timing input missing	Check J103 and system timing source

END OF MANUAL

Legacy note: This manual predates the current structured-authoring project. References, applicability statements and update procedures are maintained manually and can contain duplicated or superseded information.